

Watermelon consumption increases plasma arginine concentrations in adults.



OBJECTIVE: Watermelon is a rich source of citrulline, an amino acid that can be metabolized to arginine, a conditionally essential amino acid for humans. Arginine is the nitrogenous substrate used in the synthesis of nitric oxide and plays an essential role in cardiovascular and immune functions. No detailed studies have been conducted to evaluate plasma arginine response in humans after long-term feeding of citrulline from natural plant sources. This study investigated if watermelon juice consumption increases fasting concentrations of plasma arginine, ornithine, and citrulline in healthy adult humans. METHODS: Subjects (n = 12-23/treatment) consumed a controlled diet and 0 (control), 780, or 1560 g of watermelon juice per day for 3 wk in a crossover design. The treatments provided 1 and 2 g of citrulline per day. Treatment periods were preceded by washout periods of 2 to 4 wk. RESULTS: Compared with the baseline, fasting plasma arginine concentrations increased 12% after 3 wk of the lower-dose watermelon treatment; arginine and ornithine concentrations increased 22% and 18%, respectively, after 3 wk of the higher-dose watermelon treatment. Fasting citrulline concentrations did not increase relative to the control but remained stable throughout the study. CONCLUSION: The increased fasting plasma concentrations of arginine and ornithine and stable concentrations of plasma citrulline in response to watermelon juice consumption indicated that the citrulline from this plant origin was effectively converted into arginine. These results demonstrate that plasma concentration of arginine can be increased through intake of citrulline from watermelon.